Poisson Summation Formula

An End-to-End Meta-Math Walkthrough

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Abstract

A complete authored walkthrough of poisson summation formula, with explicit mathematical contracts, local verification gates, and reusable exports. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

HA-05 / Harmonic Analysis / Report v1

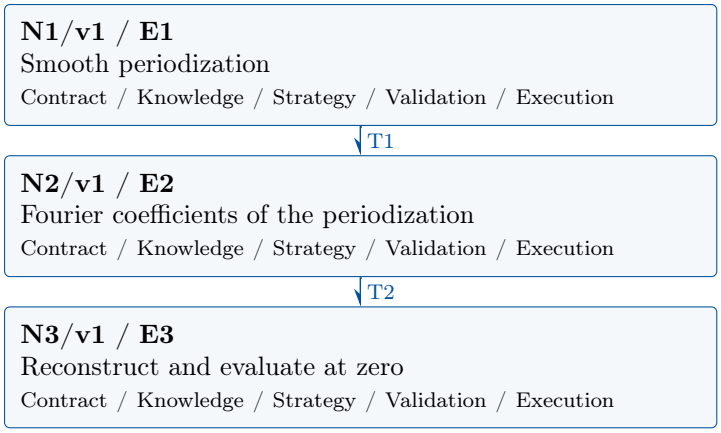
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *For $f \in \mathcal{S}(\mathbb{R})$ with HA-04's Fourier convention, prove $\sum_{n\in\mathbb{Z}} f(n) = \sum_{k\in\mathbb{Z}} \widehat{f}(k)$.*

A. Foundations and permitted imports

Use Schwartz rapid-decay bounds for f and its derivatives. Import uniqueness of Fourier coefficients for continuous periodic functions from HA-03, scaled to period1; if a continuous function has all coefficients zero, L^2 density makes it zero everywhere. Only absolutely convergent integrals and sums are interchanged.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1: N1/E1/v1 to N2. A smooth periodic function P with absolutely convergent periodization.

Receiving activity. N2: Fourier coefficients of the periodization

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Uniform summability on[0,1]; prior exports E1/v1

T2: N2/E2/v1 to N3. The kth coefficient of P is $\widehat{f}(k)$.

Receiving activity. N3: Reconstruct and evaluate at zero

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Smooth P and identified coefficients; prior exports E2/v1

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Smooth periodization

Object and role. Use rapid decay uniformly on a fundamental interval.

Inputs. $f \in S(R)$; foundational imports stated in §1

C — Contract

Own E1/v1: A smooth periodic function P with absolutely convergent periodization.

K — Knowledge

Retain $f \in S(R)$; foundational imports stated in §1; keep their assumptions and versions attached.

S — Strategy

Use rapid decay uniformly on a fundamental interval.

V — Validation

Decay estimates must be uniform in x, not merely pointwise.

E — Execution

Carry out the numbered deductions and publish E1/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N1.1 For $x \in [0,1]$ and each derivative orderr, $|f^{(r)}(x+n)| \le C_r(1+|n|)^{(-2)}$ after enlarging the constant for finitely many n.

N1.2 Hence $P(x) = \sum_n f(x+n)$ and every derivative series converge uniformly absolutely.

N1.3 Termwise differentiation shows P is smooth; shifting the integer index shows P is 1-periodic.

Export E1/v1. A smooth periodic function P with absolutely convergent periodization.

Acceptance gate. Decay estimates must be uniform in x, not merely pointwise. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N1: Decay estimates must be uniform in x, not merely pointwise. Recheck dependent nodes N2, N3.

B. N2/v1: Fourier coefficients of the periodization

Object and role. Unfold the integral into all translated intervals.

Inputs. Uniform summability on[0,1]; prior exports E1/v1

C — Contract

Own E2/v1: The kth coefficient of P is $\widehat{f}(k)$.

K — Knowledge

Retain Uniform summability on[0,1]; prior exports E1/v1; keep their assumptions and versions attached.

S — Strategy

Unfold the integral into all translated intervals.

V — Validation

The integer k is crucial for the phase cancellation.

E — Execution

Carry out the numbered deductions and publish E2/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N2.1 For $k \in Z$, its Fourier coefficient is $\int_0^1 P(x)e^{(-2\pi ikx)}dx = \sum_n \int_0^1 f(x+n)e^{(-2\pi ikx)}dx$.

N2.2 Substitute $u = x + n$.

N2.3 Since $e^{(2\pi ikn)} = 1$, the sum of integrals over[n,n+1] equals $\int_R f(u)e^{(-2\pi iku)}du = \widehat{f}(k)$.

N2.4 All rearrangements follow from absolute integrability.

Export E2/v1. The kth coefficient of P is $\widehat{f}(k)$.

Acceptance gate. The integer k is crucial for the phase cancellation. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N2: The integer k is crucial for the phase cancellation. Recheck dependent nodes N3.

C. N3/v1: Reconstruct and evaluate at zero

Object and role. Prove absolute Fourier convergence before point evaluation.

Inputs. Smooth P and identified coefficients; prior exports E2/v1

C — Contract

Own E3/v1: Poisson summation for Schwartz functions with the stated Fourier convention.

K — Knowledge

Retain Smooth P and identified coefficients; prior exports E2/v1; keep their assumptions and versions attached.

S — Strategy

Prove absolute Fourier convergence before point evaluation.

V — Validation

An L^2 identity alone would not license evaluation at a chosen point.

E — Execution

Carry out the numbered deductions and publish E3/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N3.1 Integrating by parts twice in the coefficient formula for periodic P, with matching endpoint derivatives, gives $|\widehat{f}(k)| \le \|P''\|_1/(4\pi^2k^2)$ for $k \neq 0$.

N3.2 Thus $Q(x) = \sum_k \widehat{f}(k)e^{(2\pi ikx)}$ converges uniformly absolutely and is continuous.

N3.3 Its coefficients equal those of P by termwise integration.

N3.4 Fourier uniqueness makes $P = Q$.

N3.5 $Atx = 0$ this gives $\sum_n f(n) = P(0) = Q(0) = \sum_k \widehat{f}(k)$.

N3.6 Both numerical series are absolutely convergent.

Export E3/v1. Poisson summation for Schwartz functions with the stated Fourier convention.

Acceptance gate. An L^2 identity alone would not license evaluation at a chosen point. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N3: An L^2 identity alone would not license evaluation at a chosen point. Withhold this terminal result until its gate passes.

IV. CONTINUOUS MATHEMATICAL PROOF

Read these steps in order as one proof; the diagram is optional. Numbered activities identify the specific constructions and deductions owned by each node.

A. §4.N1 — Smooth periodization

1. For $x \in [0,1]$ and each derivative orderr, $|f^{(r)}(x+n)| \le C_r(1+|n|)^{(-2)}$ after enlarging the constant for finitely many n.
2. Hence $P(x) = \sum_n f(x+n)$ and every derivative series converge uniformly absolutely.
3. Termwise differentiation shows P is smooth; shifting the integer index shows P is 1-periodic.

B. §4.N2 — Fourier coefficients of the periodization

1. For $k \in Z$, its Fourier coefficient is $\int_0^1 P(x)e^{(-2\pi ikx)}dx = \sum_n \int_0^1 f(x+n)e^{(-2\pi ikx)}dx$.
2. Substitute $u = x+n$.
3. Since $e^{(2\pi ikn)} = 1$, the sum of integrals over[n,n+1] equals $\int_R f(u)e^{(-2\pi iku)}du = \widehat{f}(k)$.
4. All rearrangements follow from absolute integrability.

C. §4.N3 — Reconstruct and evaluate at zero

1. Integrating by parts twice in the coefficient formula for periodic P, with matching endpoint derivatives, gives $|\widehat{f}(k)| \le \|P''\|_1/(4\pi^2k^2)$ for $k \ne 0$.
2. Thus $Q(x) = \sum_k \widehat{f}(k)e^{(2\pi ikx)}$ converges uniformly absolutely and is continuous.
3. Its coefficients equal those of P by termwise integration.
4. Fourier uniqueness makes $P = Q$.
5. $Atx = 0$ this gives $\sum_n f(n) = P(0) = Q(0) = \sum_k \widehat{f}(k)$.
6. Both numerical series are absolutely convergent.

□

V. WORKED ORCHESTRATION

For $f(x) = e^{(} - \pi t\, x^2)$, $t > 0$, the Gaussian transform is $t^{(-1/2)}e^{(-\pi\xi^2/t)}$. Thus $\sum_n e^{(} - \pi t\, n^2) = t^{(-1/2)} \sum_k e^{(-\pi k^2/t)}$. At $t = 1$ the two series agree termwise; at other t the formula relates different Gaussian scales.

A. Boundary and hypothesis audit

Changing a function only at integers leaves its Lebesgue Fourier transform unchanged but can change the left-hand sampled sum. This shows why arbitraryL ^{1} equivalence classes cannot satisfy a point-sampling formula without additional regularity. Validation retains the Schwartz scope.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Failure and correction

This is an authored teaching trace, not a recorded agent run.

1. **Contract → Knowledge:** register the §1 theorem, imports, and requested exports before beginning the proof.
2. **Strategy → Execution:** choose the bridges in §3 and carry out the numbered activities in §4; transmit only established E-exports along the §2 DAG.
3. **Validation → responsible node:** test the specific boundary/failure in §5 and the node's individual audit. A failing candidate is withheld; downstream nodes may not use it.
4. **Knowledge + Strategy:** identify whether the error is an omitted condition, an invalid inference, or an attempted extension beyond the theorem. Restore the exact contract or explicitly state a different one.
5. **Execution:** redo the affected mathematical activity using the repaired condition or inference. Recheck consuming nodes, rather than feeding their conclusions back as premises.
6. **Contract:** publish only the supported export with its assumptions and proof location. A hypothetical revised export is v2; the written proof here is the v1 exposition, not a claim that this repair was executed.

B. Hypothesis boundary

Changing a function only at integers leaves its Lebesgue Fourier transform unchanged but can change the left-hand sampled sum. This shows why arbitraryL ^{1} equivalence classes cannot satisfy a point-sampling formula without additional regularity. Validation retains the Schwartz scope.

C. Version and recheck

Treat any changed assumption as a new contract. Recheck the revised export and all of its descendants. A backward diagnostic never establishes an earlier theorem. This is an authored teaching trace, not an execution log.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.